

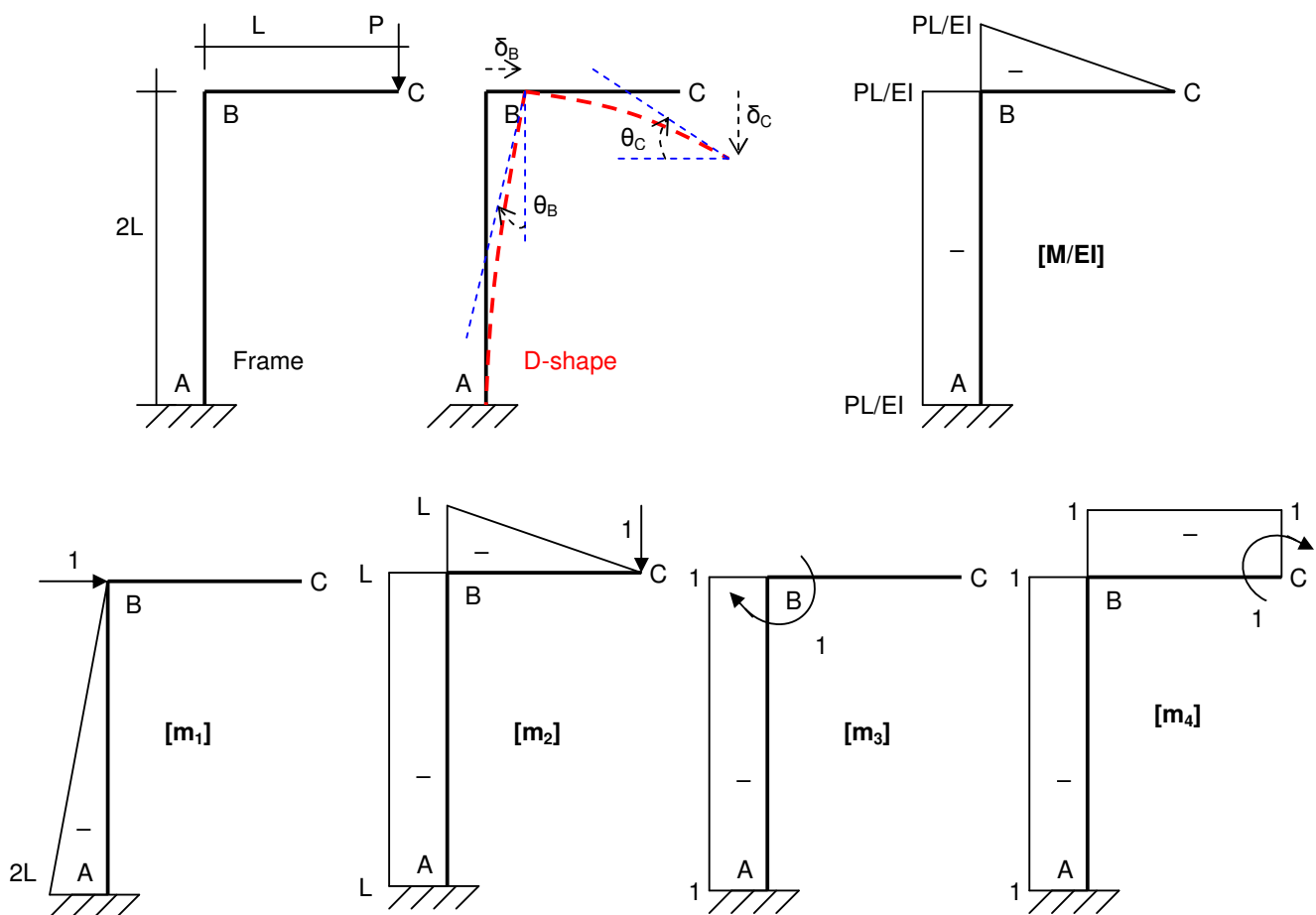
4.4 FRAME DISPLACEMENTS

The Virtual Unit Load method can easily be applied to determine the displacements in rigid-jointed frames. The solution procedure is as outlined in article 4.2.3, if we replace the word beam with the word frame. Each member longitudinal axis is used as an x-axis to express m_i and M/EI over the whole length of the member. Only flexural strains are considered. Examples 1 and 2 below illustrate the technique. ALT 2 shall be preferred.

Example 1

Sketch the D-shape and determine $\delta_B(\rightarrow)$, $\delta_C(\downarrow)$, θ_B and θ_C for the frame shown below. Assume $EI = \text{constant}$.

Solution



Calculations

Let $\delta_B(\rightarrow) = \delta_1$, $\delta_C(\downarrow) = \delta_2$, $\theta_B = \delta_3$ and $\theta_C = \delta_4$

$$\delta_B(\rightarrow) = \delta_1 = \int_L m_1 (M/EI) dx = (1/2)acL = (1/2)(-2L)(-PL/EI)(2L) = \mathbf{2PL^3/EI}$$

$$\begin{aligned}\delta_C(\downarrow) = \delta_2 &= \int_L m_2 (M/EI) dx = acL + (1/3)acL = (-L)(-PL/EI)(2L) + (1/3)(-L)(-PL/EI)(L) \\ &= \mathbf{7PL^3/3EI}\end{aligned}$$

$$\theta_B = \delta_3 = \int_L m_3 (M/EI) dx = acL = (-1)(-PL/EI)(2L) = \mathbf{2PL^2/EI}$$

$$\begin{aligned}\theta_C = \delta_4 &= \int_L m_4 (M/EI) dx = acL + (1/2)acL = (-1)(-PL/EI)(2L) + (1/2)(-1)(-PL/EI)(L) \\ &= \mathbf{5PL^2/2EI}\end{aligned}$$

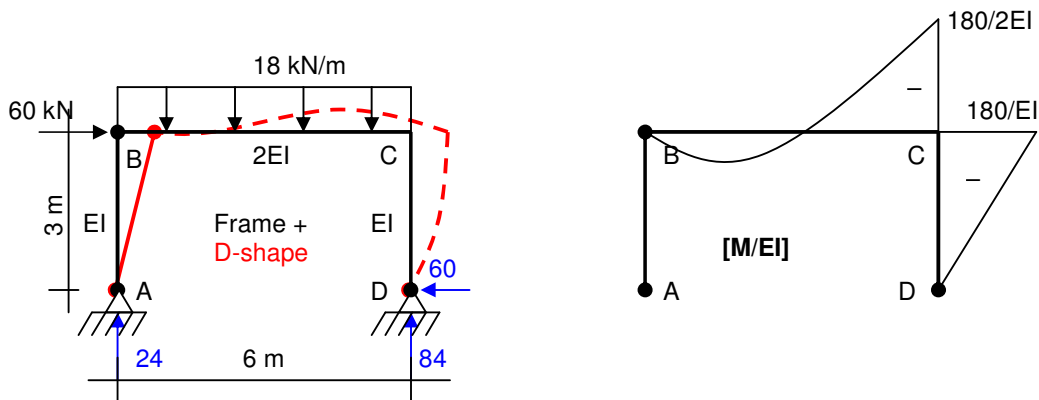
Note

Because axial strains are negligible, $\delta_C(\rightarrow) = \delta_B(\rightarrow) = 2PL^3/EI$.

This horizontal deflection is called the **sway**

Example 2

Determine $\delta_B(\rightarrow)$, θ_A , θ_B , θ_C and θ_D for the frame shown below. Take $EI = 55.8 \times 10^3 \text{ kNm}^2$.



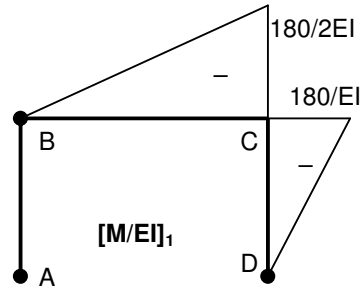
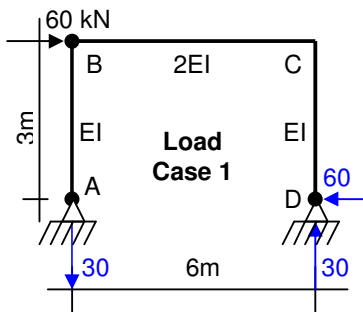
NOTE

The M/EI diag over BC is NOT a standard shape and hence ALT 2 CANNOT be used directly. This problem is solved by decomposing the load into two cases and using superposition.

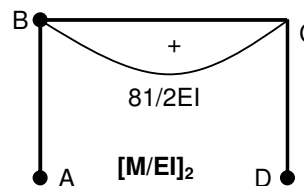
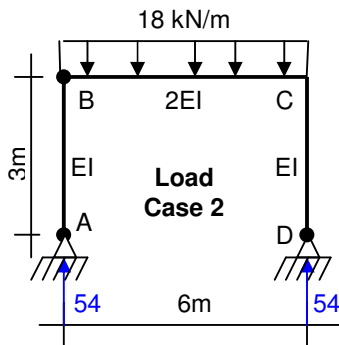
Solution by Superposition

Let $\delta_B(\rightarrow) = \delta_1$, $\theta_A = \delta_2$, $\theta_B = \delta_3$, $\theta_C = \delta_4$ and $\theta_D = \delta_5$.

Load case 1: 60 kN only

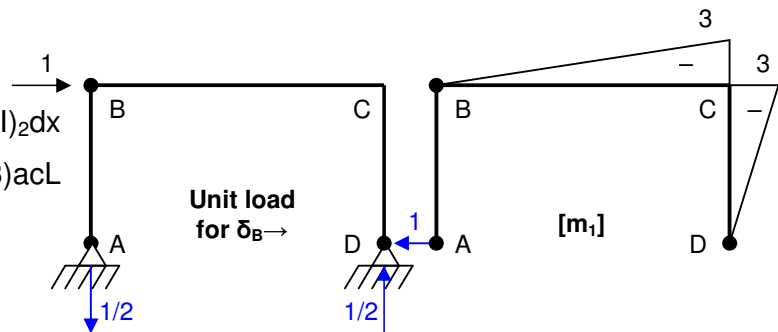


Load case 2: 18 kN/m udl only

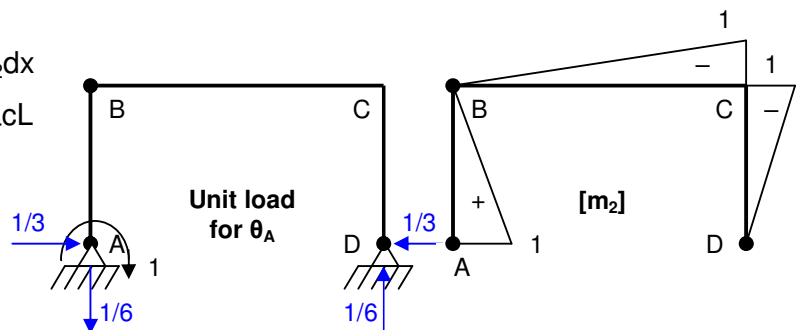


Calculations

$$\begin{aligned}
 \delta_{B \rightarrow} &= \int_L m_1 (M/EI)_1 dx + \int_L m_1 (M/EI)_2 dx \\
 &= (1/3)acL + (1/3)acL + (1/3)acL \\
 &= (1/3)(-3)(-90/EI)(6) \\
 &\quad + (1/3)(-3)(-180/EI)(3) \\
 &\quad + (1/3)(-3)(81/2EI)(6) \\
 &= 837/EI = 15 \text{ mm}
 \end{aligned}$$

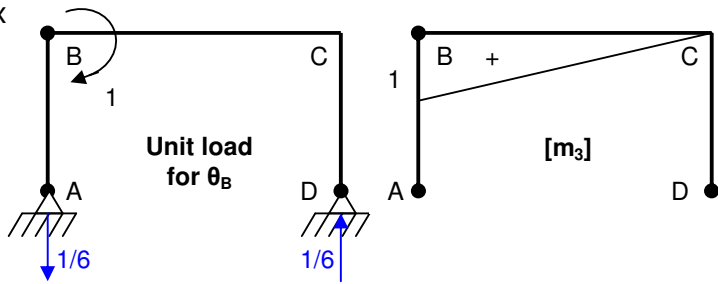


$$\begin{aligned}
 \theta_A &= \int_L m_2 (M/EI)_1 dx + \int_L m_2 (M/EI)_2 dx \\
 &= (1/3)acL + (1/3)acL + (1/3)acL \\
 &= (1/3)(-1)(-90/EI)(6) \\
 &\quad + (1/3)(-1)(-180/EI)(3) \\
 &\quad + (1/3)(-1)(81/2EI)(6) \\
 &= 279/EI = 5 \times 10^{-3} \text{ rad}
 \end{aligned}$$

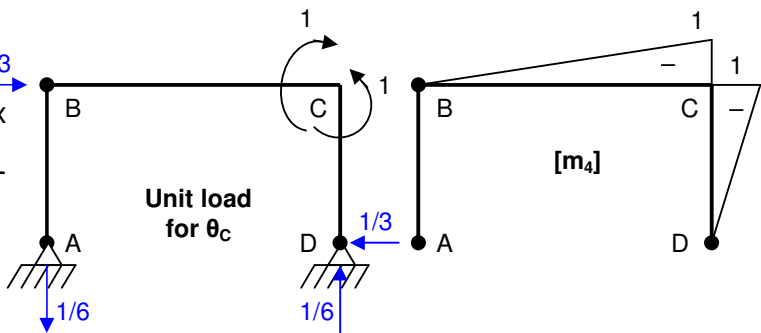


[ALT: $\theta_A = \delta_B/3 = 279/EI$]

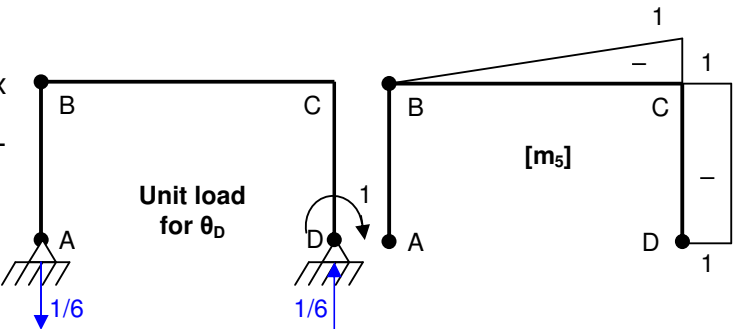
$$\begin{aligned}
 \theta_B &= \int_L m_3(M/EI)_1 dx + \int_L m_3(M/EI)_2 dx \\
 &= (1/6)acL + (1/3)acL \\
 &= (1/6)(1)(-90/EI)(6) \\
 &\quad + (1/3)(1)(81/2EI)(6) \\
 &= -9/EI = -0.16 \times 10^{-3} \text{ rad}
 \end{aligned}$$



$$\begin{aligned}
 \theta_C &= \int_L m_4(M/EI)_1 dx + \int_L m_4(M/EI)_2 dx \\
 &= (1/3)acL + (1/3)acL + (1/3)acL \\
 &= (1/3)(-1)(-90/EI)(6) \\
 &\quad + (1/3)(-1)(-180/EI)(3) \\
 &\quad + (1/3)(-1)(81/2EI)(6) \\
 &= 279/EI = 5 \times 10^{-3} \text{ rad}
 \end{aligned}$$



$$\begin{aligned}
 \theta_D &= \int_L m_5(M/EI)_1 dx + \int_L m_5(M/EI)_2 dx \\
 &= (1/3)acL + (1/2)acL + (1/3)acL \\
 &= (1/3)(-1)(-90/EI)(6) \\
 &\quad + (1/2)(-1)(-180/EI)(3) \\
 &\quad + (1/3)(-1)(81/2EI)(6) \\
 &= 369/EI = 6.61 \times 10^{-3} \text{ rad}
 \end{aligned}$$



4.5 HYBRID FRAMES

4.5.1 Description

Hybrid frames are frames which have both pin-jointed members (truss type) and rigid-jointed members (beam or frame type). The pin-jointed members suffer only axial strains. The rigid-jointed members suffer flexural, shear and axial strains. However, for purposes of calculating displacements, axial strains in the pin-jointed members, and only the flexural strains in the beam or rigid-jointed members shall be considered.

4.5.2 Calculation of Displacements

The displacements in hybrid frames are calculated using a combination of equation (2*) of article 4.2.2 and equation (4*) of article 4.4.3 to obtain the equation (5*).

Thus: -

$$\delta_i = \int_L m_i (M/EI) dx + \sum n_i (N/EA) L \quad \text{-----} \quad \text{Eqn 5*}$$

4.5.3 Solution Procedure

The solution procedure is therefore a combination of the procedures outlined in articles 4.2.3 and 4.4.4, thus:

1. Analyse the hybrid frame subject to the given loading to obtain **[M/EI]** diagram and **[N]** member forces.
2. For any required **ith** displacement: -
 - ✓ Assume the applied loading was absent
 - ✓ Apply a virtual unit load at the point of interest, in the direction and sense of interest
 - ✓ Analyse the structure to obtain the **[m_i]** diagram and **[n_i]** member forces
3. Next required displacement.
4. Make a suitable table to facilitate your calculations.
5. For each required displacement, evaluate
$$\delta_i = \int_L m_i (M/EI) dx + \sum n_i (P/EA) L$$
6. Next required displacement
7. Stop: END.

Example 1

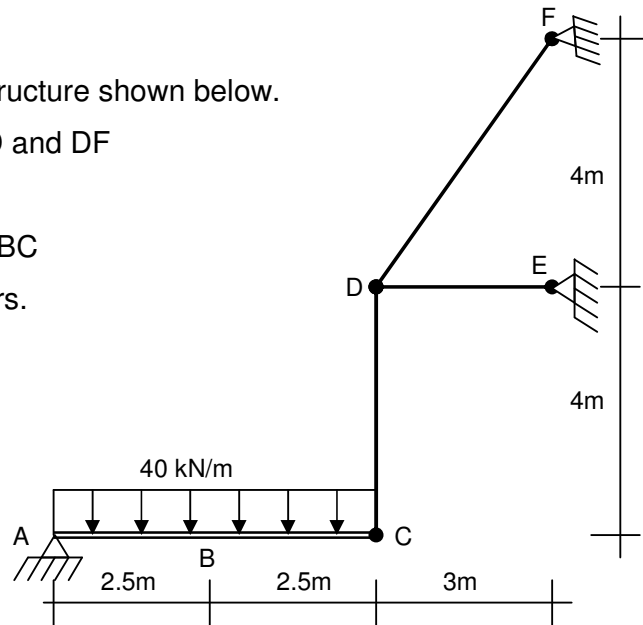
Determine $\delta_B(\downarrow)$ and $\delta_C(\downarrow)$ for the structure shown below.

Take $A = 600 \text{ mm}^2$ for members CD and DF

$A = 500 \text{ mm}^2$ for member DE

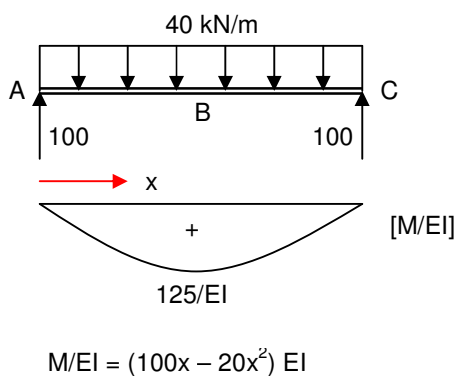
$I = 84 \times 10^6 \text{ mm}^4$ for member ABC

And $E = 200 \text{ kN/mm}^2$ for all members.



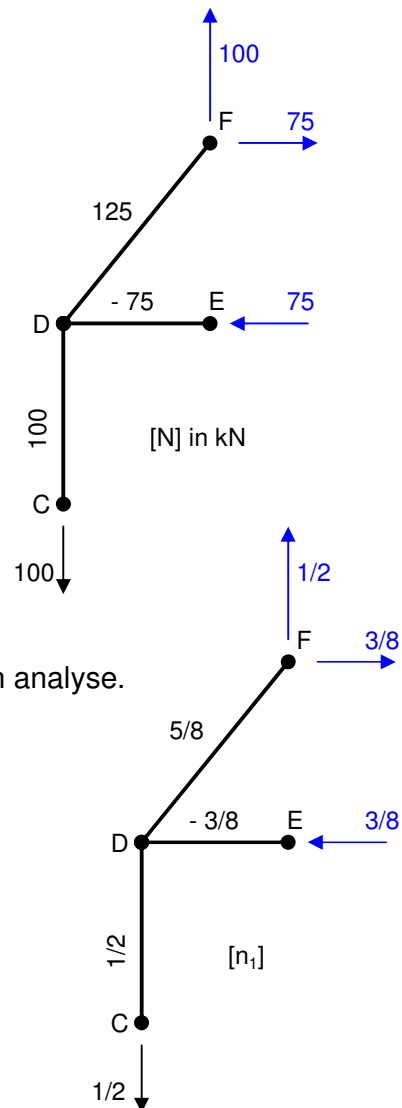
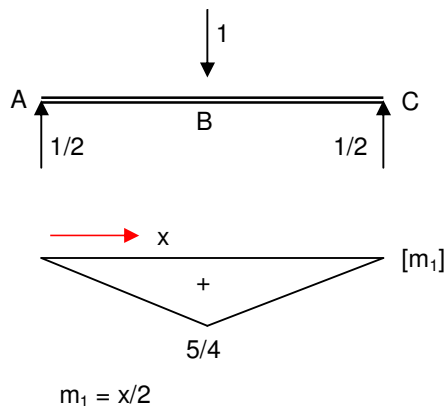
Solution

Isolate the beam ABC from the truss, then analyse.



Let $\delta_B(\downarrow) = \delta_1$

Apply a virtual unit load vertically downwards at B, then analyse.



Let $\delta_C(\downarrow) = \delta_2$.

Apply a virtual unit load vertically downwards at C, then analyse.

[Notice that such a load will be taken by the truss and the beam will suffer no flexure]

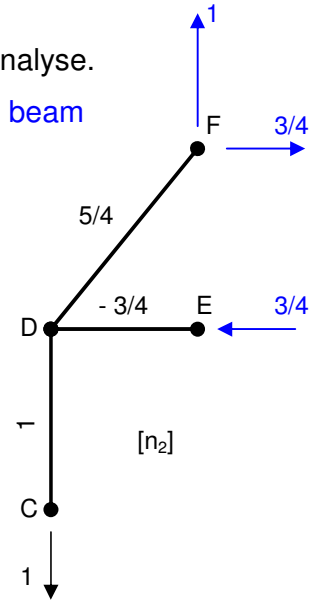
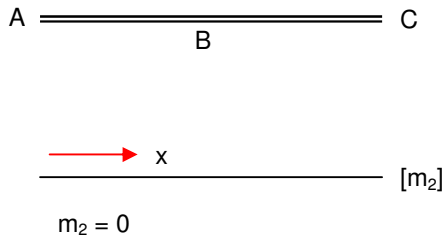


Table 1: Flexural Strains

Member	L (m)	M/EI	m_1	m_2	Limits (m)	$\int_L m_1 (M/EI) dx$	$\int_L m_2 (M/EI) dx$
AB	2.5	$(100x - 20x^2)/EI$	$x/2$	0	$0 \leq x \leq 2.5$	$15625/96EI$	0
CB	2.5	$(100x - 20x^2)/EI$	$x/2$	0	$0 \leq x \leq 2.5$	$15625/96EI$	0

Table 2: Axial Strains

Member	L(m)	$EAx10^3$ (kN)	N (kN)	n_1	n_2	$n_1NL/EAx10^{-3}$ (m)	$n_2NL/EAx10^{-3}$ (m)
CD	4	120	100	1/2	1	1.6667	3.3333
DE	3	100	- 75	- 3/8	- 3/4	0.8438	1.6875
DF	5	120	125	5/8	5/4	3.2552	6.5104
Σ						5.7657	11.5312

Calculations

Given $EI = 200 \text{ kN/mm}^2 \times 84 \times 10^6 \text{ mm}^4 = 16.8 \times 10^9 \text{ kN mm}^2 = 16.8 \times 10^3 \text{ kNm}^2$

$$\begin{aligned}
 \delta_B(\downarrow) &= \delta_1 = \int_L m_1 (M/EI) dx + \sum n_1 (N/EA) L = 2(15625/96EI) + 5.7657 \times 10^{-3} \\
 &= 15625/48EI + 5.7657 \times 10^{-3} \\
 &= 19.3762 \times 10^{-3} + 5.7657 \times 10^{-3} \\
 &= 25.1 \times 10^{-3} \text{ m} \\
 &= \mathbf{25.1 \text{ mm}}
 \end{aligned}$$

$$\begin{aligned}
 \delta_C(\downarrow) &= \delta_2 = \int_L m_2 (M/EI) dx + \sum n_2 (N/EA) L = 0 + 11.5312 \times 10^{-3} \\
 &= 11.5 \times 10^{-3} \text{ m} \\
 &= \mathbf{11.5 \text{ mm}}
 \end{aligned}$$